

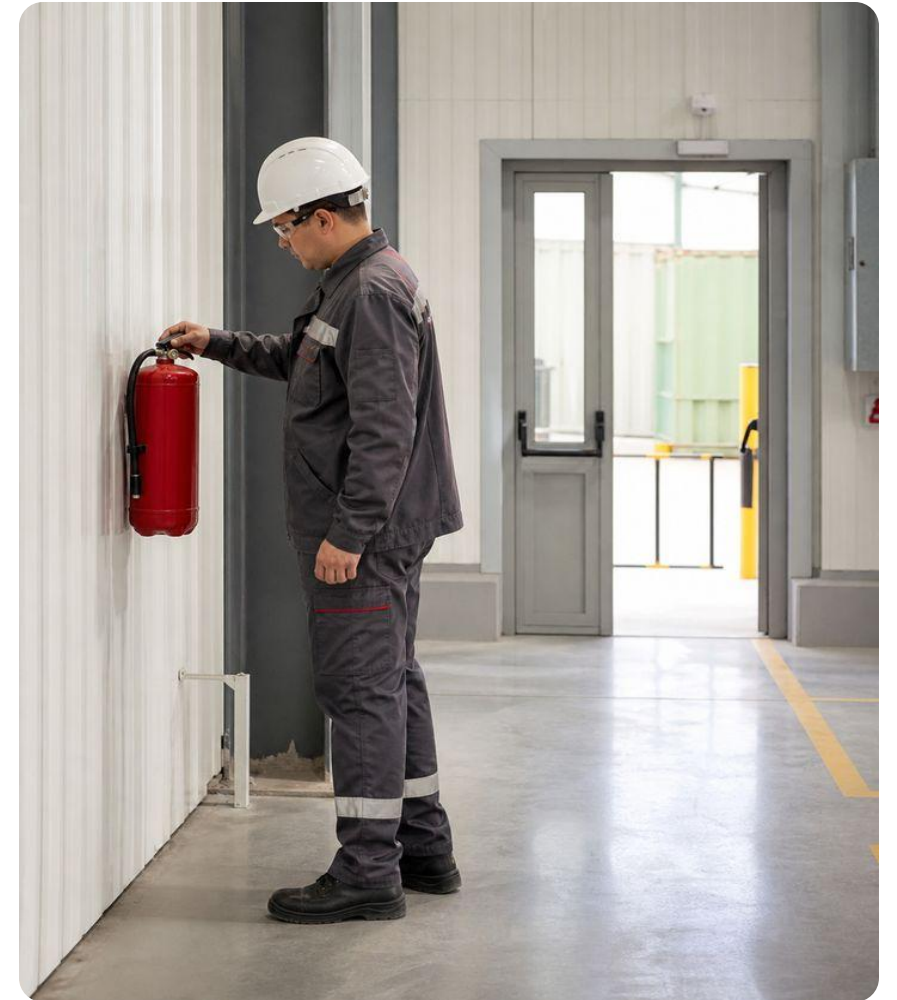


TRAINING COURSE

fire safety

Fire prevention, evacuation and safe use of fire extinguisher

Training course for workers and line personnel





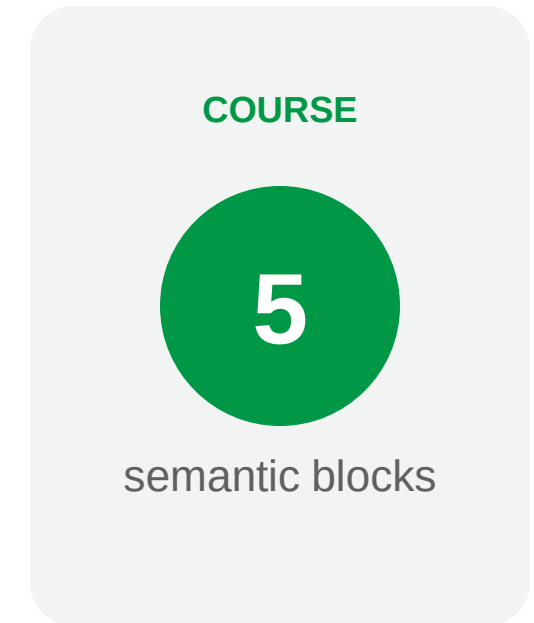
After the course worker will be able to

- 01 — Name the first actions in case of smoke, burning smell, alarm or open fire.
- 02 — Understand fire conditions, hazards, and methods to stop a fire.
- 03 — Select the appropriate fire extinguisher according to the marking and evaluate the safety of extinguishing.
- 04 — Evacuate, inform 101/112 and act at the collection point.

Life and safety evacuation take precedence over property and extinguishing.

| Course logic

- 01 — First actions and basics of combustion.
- 02 — Fire classes and choice of fire extinguisher.
- 03 — evacuation, message and collection point.
- 04 — Prevention and hot work.
- 05 — Practical scenarios and final algorithm.

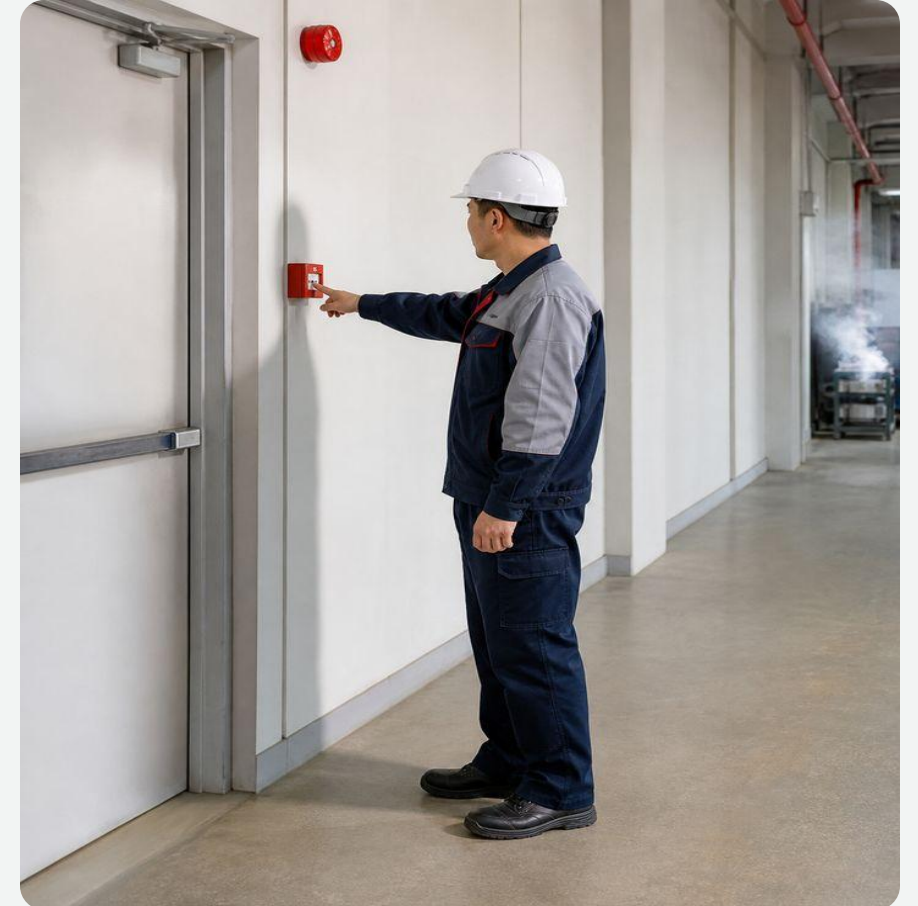




01

Signal and combustion basics

If there are signs of fire, first warn people, inform them and begin evacuating.





Alarm requires immediate action

1

Report

Signal and call 101 or 112 if required by role.

2

Warn

Notify people nearby and follow the instructions of the warning system.

3

Evacuate

Follow a safe route without stopping for things.

4

Get ready

Arrive at the collection point, go through roll call and do not return.

The dangerous phrase “do not raise the alarm” has been removed: the alarm is raised immediately.

| Fire - uncontrolled burning

- A fire threatens people, property, operations and the environment.
- Its development depends on fuel, oxygen, ignition energy and conditions in the room.
- A small fire can quickly grow beyond the stage suitable for manual firefighting.
- Early detection and reporting shorten the period of uncontrolled fire development.

Flame size is not the only indicator of danger; smoke may be more critical.



Hazardous factors act before the flame

risk / SOURCE

CONSEQUENCE

CONTROL MEASURE

Smoke and toxic products

Poisoning, disorientation

Early warning and evacuation

High temperature

Burns and heat exposure

Distance and protected route

Reduced oxygen

Loss of consciousness

Do not enter without special protection

Collapse and explosion

Injuries and blocked exit

Do not approach and inform the authorities

Do not return to a smoky room without special powers and protection.

Reasons related to source control

- Faulty electrical equipment, overload and damaged insulation.
- Open flame, sparks and hot work without preparation.
- Improper storage of flammable materials and liquid spills.
- Smoking outside the designated area, hot surfaces and poor cleaning.



Prevention controls the ignition source and fire load simultaneously.

| Three conditions are required for combustion

- Combustible substance - a material, vapor or gas capable of supporting combustion.
- The oxidizing agent is usually oxygen from the air.
- Ignition source - flame, spark, hot surface or other energy.
- Control of any element can prevent the occurrence or development of a fire.

The fire triangle helps to choose a measure of prevention and extinguishing.

| Heat travels in four ways

- Conduction transfers heat through the material and connected structures.
- Convection carries hot gases and smoke upward and throughout rooms.
- Radiation heats objects at a distance.
- The flame directly ignites materials upon contact.

**A closed door
can slow the
spread of smoke
and heat.**



Combustion is stopped by affecting the condition

1

Cool

Reduce temperature with a suitable extinguishing agent.

2

Isolate

Restrict oxygen access or cover surface.

3

Remove fuel

Stop feed or remove material if safe to do so.

4

Abort reaction

Apply a suitable composition for the fire class.

The wrong product can spread the fire or create a new hazard.



02

Classes and fire extinguishers

The choice is made according to the burning material, the presence of voltage and the marking of a particular fire extinguisher.

| The fire class determines the fuel

- A - hard materials, usually forming smoldering coals.
- B - flammable liquids and melting solids.
- C - flammable gases; D - combustible metals.
- For live electrical equipment, take into account the electrical risk and the marking of the product; Class F applies to edible oils and fats if it is provided for the facility.

Case color or foreign label does not replace the class designation and instructions.



Voltage limits the choice of remedy



- If possible, safely de-energize equipment according to established procedure.
- Do not apply water or foam to live equipment unless the product is specifically designed for use.
- Observe the fire extinguisher markings and the manufacturer's safety distances.
- After relieving the tension, consider the class of the material actually burning.

The equipment switch does not always guarantee the absence of voltage.

| Choose the Fire Extinguisher for the Task

- Water cools and applies only to suitable materials and conditions.
- Foam insulates the surface of a flammable liquid, but has electrical hazard limitations.
- Powder is universal in many classes, but reduces visibility and contaminates equipment.
- Carbon dioxide does not leave a residue, but requires consideration of suffocation, cold bell and re-ignition.

Choose according to the label of a specific fire extinguisher, and not according to habit.



| Each type has limitations

risk / SOURCE

CONSEQUENCE

CONTROL MEASURE

Water

Electric risk; reaction with individual substances

Use only as labeled and after evaluation

Foam

Conductivity and incompatibility with certain fluids

Do not use outside the designated area

Powder

Dust, loss of visibility, re-ignition

Maintain exit and control the outbreak

CO₂

Cold burn, oxygen displacement

Do not touch the discharge horn; ventilate the area safely

Fire extinguishing ability does not negate the danger of the extinguishing method itself.

| Marking contains a solution for the employee

- Check the type of charge and the designated fire classes.
- Read voltage, temperature and application restrictions.
- Make sure that the instructions are readable, the seal is intact and the service period is valid.
- If the labeling is unclear, do not use the product at random.

The old foreign color scheme is not used as a mandatory marking for Kazakhstan.



External verification takes a few seconds

1

Access

fire extinguisher is visible,
marked and unobstructed.

2

Status

The body, hose, discharge
horn and handle are free
from damage and corrosion.

3

Ready

The seal and receipt are
intact; pressure indicator in
the permissible zone, if
available.

4

Maintenance

Marking and inspection mark
are legible and valid.

Damaged fire extinguisher is not checked by starting indoors.

| Accommodation depends on room category

- Maximum distance to a possible fire: 20 m in public buildings.
- For categories A, B and V1–V4 — 30 m; for category G — 40 m; for category D — 70 m.
- Extinguishers are placed in visible locations and near evacuation exits without obstructing evacuation.
- The top of an extinguisher must be no higher than 1.5 m; when its total mass is 15 kg or more, no higher than 1.0 m.

The universal “30 m and 1.1 m” from the old material were erroneous.



| First decide: put out or leave



- Firefighting is permitted only for a small incipient fire and only if you have been trained.
- A serviceable extinguisher suitable for the fire and a safe retreat route behind you must be available.
- There must be no dense smoke, explosion hazard, toxic release or rapid fire spread.
- If any condition is not met, warn people, evacuate and notify the emergency services.

fire extinguisher does not oblige you to risk your life.



fire extinguisher are used steadily and precisely

1

Prepare

From a safe position, remove the seal and pull out the safety pin.

2

Send

Hold the nozzle at the base of the flame, not into the smoke.

3

Submit

Press the lever and dispense the mixture briefly or continuously according to the instructions.

4

Move

Fight the fire systematically while keeping the exit clear and monitoring conditions.

The specific sequence is specified by the marking of the fire extinguisher.

| The escape route remains behind

- Approach only as far as the instructions and thermal exposure allow.
- Do not walk into smoke or allow fire to cut off your exit.
- Consider the direction of smoke and wind, do not stand in the emission of combustion products.
- Retreat immediately if the flame grows, visibility deteriorates, or the charge runs out.

The position of the worker is more important than trying to get as close to the fire as possible.



Extinguishing is stopped when control is lost



- Fire spreads beyond the initial seat of fire or upward along the structure.
- Smoke becomes dense, visibility deteriorates or the retreat route changes.
- Gas, explosion, hazardous substances or concealed combustion are suspected.
- The extinguisher does not operate, its extinguishing agent is exhausted or the wrong extinguisher was selected.

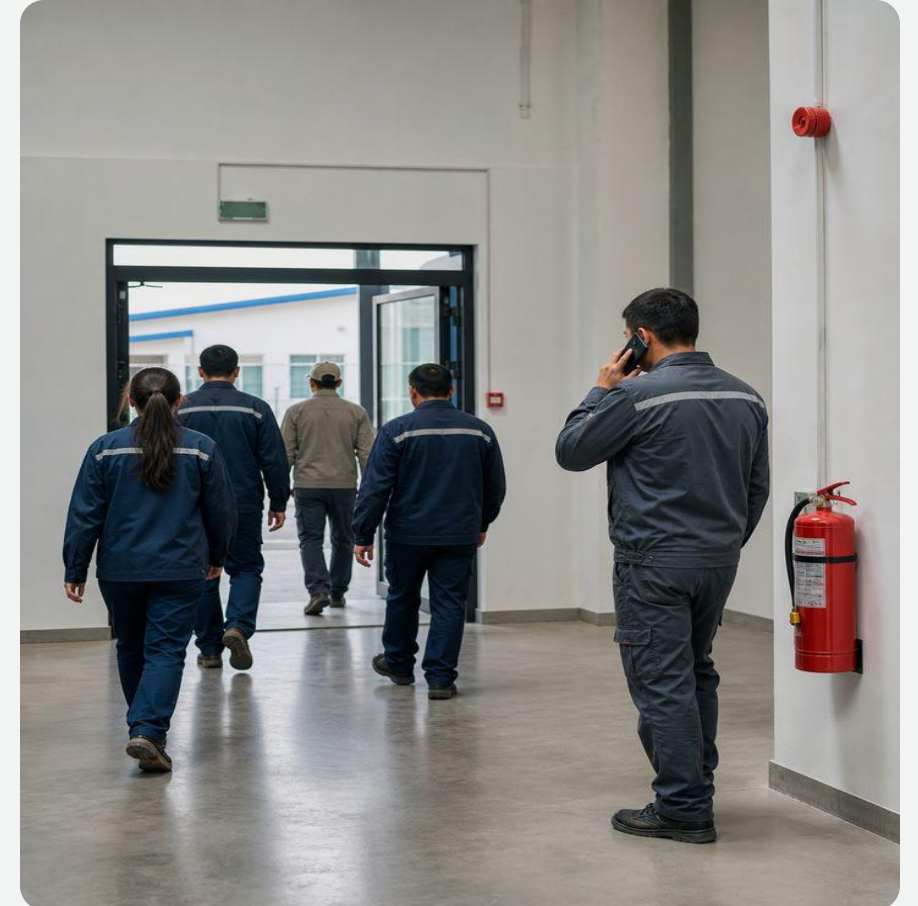
Retreating on time is a professional decision, not a failure.



03

evacuation and message

The route, communication and accounting of people must be clear before a fire occurs.



| Evacuation routes are always clear

- Passages, stairs, doors and exits are not cluttered with material and equipment.
- Doors on the escape route open and are not blocked by improvised devices.
- Signs and emergency lighting must be visible.
- A faulty or blocked path is reported immediately.

The escape route is checked during normal operation and not during an alarm.

You need to be able to read the evacuation plan

- Find your current position, main route and alternate route.
- Label manual call points, extinguishing media and collection point if shown.
- Check the plan against actual doors and passages.
- Changing the layout requires updating the diagram and instructions.



The plan is a route hint, but the solution takes into account actual smoke and obstacles.



When calling 101 or 112, tell the main thing

1

Address

Object, site, floor, landmark
and safe entry.

2

What's burning

Material, equipment,
premises and visible signs.

3

People & Hazards

Injured, trapped, gas,
cylinders or chemicals.

4

Contact

Name, number and who will
meet the arriving units.

Do not end the conversation before the dispatcher and answer briefly and accurately.

| Get out quickly, but don't panic

- Leave your personal belongings behind and take the nearest safe route.
- Assist in assigned role without creating cross flow.
- Close the door behind you if it is safe to do so, but do not lock it.
- The elevator is not used unless it is intended for evacuation according to a special procedure.

**A closed,
unlocked door
can slow the
spread of smoke
and fire.**

| A collection point is needed to count people

- Arrive at the designated location and remain with your group.
- Report yourself to the person responsible for roll call.
- Provide information about the possible whereabouts of the missing person.
- Do not leave the collection point or return to the building without permission.

Unauthorized departure may direct rescuers to an unnecessary search.



Re-entry prohibited without permission



- Do not return for documents, a phone, clothing or tools.
- Do not re-enter to search for a colleague; tell rescuers the person's possible location.
- Do not cross barriers or obstruct access for emergency units.
- Re-entry is permitted only after official authorization from the responsible person.

Even if no smoke is visible, toxic gases may remain inside and risk collapse.



04

Prevention and hot work

Fire is prevented by daily control of order, electricity, flammable materials and ignition sources.

| Good Housekeeping Reduces the Fire Load

- Combustible waste is disposed of regularly in the provided containers.
- Passages, exits, fire cabinets and fire extinguishers are not blocked.
- Spills are cleaned up according to procedure and contaminated materials are safely contained.
- Temporary storage is not converted into a permanent warehouse.

Cleaning is an element of fire prevention, and not just external order.

| Electrical fault is not masked

- Do not overload the network or use homemade extension cords or connections.
- Damaged cable, heat, odor and sparking require disconnection and reporting.
- The equipment is turned off upon completion if this is provided for in the instructions.
- Electrical panels and protective devices are kept accessible and closed.

Re-enabling a triggered protection without finding out the reason is dangerous.

| Store Flammable Liquids Under Control

- Use compatible closed containers with legible labels.
- Limit the quantity kept at the workplace to production needs and local rules.
- Prevent heating, sparks, spills and joint storage of incompatible substances.
- Prepare ventilation and emergency spill-containment equipment in advance.

**Liquid poured
into an unmarked
container creates
a double risk.**

| Smoking - only in designated area

- The location is designated and organized taking into account fire risk.
- Smoking is prohibited in warehouses, production areas and other places where it is established.
- Cigarette butts are completely extinguished and placed in the provided non-flammable container.
- Electronic smoking devices are also subject to local mode.

The absence of an open flame does not mean the absence of a heating source.



Prepare Hot Work Before Striking an Arc

1

Allow

Determine the place, permit-to-work, responsible and time.

2

Prepare

Remove or secure flammable material, isolate equipment, test environment.

3

Provide

Extinguishing media, fencing, ventilation, communications and surveillance.

4

Monitor

Monitor during and after completion in the prescribed manner.

A portable screen is not a substitute for the removal of concealed flammable materials.

| Fire Watch Continues After Work

- The fire watch monitors the spark area, concealed spaces and adjacent rooms.
- The fire watch has communications, knows how to stop the work and knows where the firefighting equipment is located.
- Fire-watch duty must not be combined with another task that distracts from monitoring.
- The duration of post-work monitoring is set by the applicable rules and local permit.

**Smoldering may
be detected after
the welder leaves.**



Smoke or a Burning Odor Calls for Action

1

Do Not Ignore

Stop the safe operation and evaluate the exit route.

2

Give signal

Warn people and report via 101/112 and local scheme.

3

Evacuate

Drive in a safe manner by closing doors without locking them.

4

Send information

State exact location, signs and possible hazardous materials.

Do not wait for visible flame or repeated signal.

Scenario: a basket at the exit caught fire

SCRIPT

- Assess the situation: the fire is small, but it is close to an evacuation route.
- Raise the alarm and help people leave before the fire can cut off the doorway.
- Attempt extinguishing only if you are trained, have a suitable fire extinguisher and have a clear escape route.
- If smoke increases, withdraw immediately, close the door without locking it and proceed to the assembly point.

The exit should not be between the worker and the fire.

Scenario: Smoke from electrical panel

SCRIPT

- Do not open the cabinet without authority and do not touch the cabinet if voltage is suspected.
- Sound the alarm, notify 101/112 and arrange evacuation.
- De-energization is carried out by an authorized person following a safe procedure.
- The use of a fire extinguisher is only permitted if it is properly marked and has a safe escape route.

Latent electric and arc hazard may be higher than visible flame.



Final algorithm in case of fire

- ✓ Report: alarm, 101 or 112, exact location and dangers.
- ✓ Warn: people nearby and responsible according to the local scheme.
- ✓ Evacuate: safe route, close door without locking.
- ✓ Gather: roll call, information about absentees, prohibition of re-entry.
- ✓ Extinguish only the safe initial fire when all conditions are met.

Click “Start Test”

Updated: 08/21/2026